

Fujitsu Quantum Simulator Challenge 2025-26 Application Form

1. Names of all Applicants' organizations (if multiple companies and/or institutions will be participating, please list all.)

Tokyo International University (TIU), E-Track Program, Digital Business and Innovation Faculty.

2. Names of all participating Members (if multiple Members will be participating, please list all.)

MD MUJAHIDUL ISLAM (Lead Developer, 3rd-year undergraduate), Gayan P. Hettiarachchi, PhD(Principal Investigator, Assistant Professor.

3. Website(s) of Applicant(s) (if multiple companies and/or institutions will be participating, please list all)

<https://www.tiu.ac.jp/etrack/>

<https://www.tiu.ac.jp/etrack/faculty/gayan-p-hettiarachchi/>

4. Mailing Address(es), including country, of all Applicants

Ikebukuro Campus, 4-42-31 Higashi-Ikebukuro, Toshima, Tokyo, 170-0013 Japan
Tel : +81-(0)3-5992-5555

5. Name and E-mail address of primary contact(s)

Primary: Gayan P. Hettiarachchi, PhD (ghettiar@tiu.ac.jp)

Technical Lead: MD MUJAHIDUL ISLAM (mujahid168216@gmail.com)

6. Describe the level of experience by Member(s) to quantum computing?

MD MUJAHIDUL ISLAM: Intermediate - Completed hybrid quantum-classical image classification(<https://github.com/mujahidmahfuz/Hybrid-Quantum-Classical-Image-Classifier>, quantumprogramming projects (<https://github.com/mujahidmahfuz/Quantum-Computing>). Currently building classical VRP baselines (OR-Tools, TSPLIB).

Gayan P. Hettiarachchi: Research supervision in ML/AI optimization; Ph.D. in Physics, strong HPC/algorithmic background.

7. Have the Members participated in other quantum computing industry challenges? If yes, please list them.

No prior industry quantum challenges. Team has ML research experience targeting academic publications. Also, MD MUJAHIDUL ISLAM'S review paper on Quantum generative adversarial network has been accepted for publication in "Quantum Information Processing".

8. Please describe the problem (the "Problem") that the Applicant(s) want to solve using the quantum simulator.

Logistic Routing for Tokyo SME Delivery: Vehicle Routing Problem (VRP) for Tokyo SME delivery fleets (20-50 daily stops) with realistic constraints: time windows (9AM-8PM), truck capacity (15 parcels), dynamic traffic disruptions. Classical exact solvers (OR-Tools) timeout >2min on 30+ stops; heuristics sacrifice 15-25% optimality (¥500k+ annual losses per fleet). Fujitsu's 40-qubit tensor-network simulator + QARP will solve larger VRP instances via QAOA on QUBO formulation, enabling real-time re-routing during peak traffic/accidents.

9. What are the application area(s) and example(s) of the Applicant solution(s)?

Application Area: Logistics & Supply Chain Optimization

Customer Example: Tokyo food/pharma delivery SME (10 trucks, 500 orders/day) facing Japan's "2024 Problem" driver shortage.

Solution: QAOA-optimized routes on Fujitsu simulator reduce total mileage 15-20% vs OR-Tools heuristics on 40-stop instances, visualized on Tokyo maps (Shibuya-Shinjuku-Ginza delivery).

10. Please list any future customer benefits of a solution to the Problem.

1. ¥1.2M annual savings per 10-truck fleet (fuel 12% + labor 18% reduction). 2. 22% fewer late deliveries 95% on-time rate vs 73% current. 3. 3x faster re-routing (90s vs 5min) during tr

11. Have the Member(s) tested the Problem on other quantum environments? If yes, describe the process and results.

No quantum VRP testing yet. Classical OR-Tools baseline complete: ATTr48.vrp (48 stops) solved in 12s (attached: vrp_baseline.ipynb).
Week 2-3 plan: Local Qiskit/Qulacs QAOA (20 qubits) before Fujitsu access. Prior quantum work: Hybrid quantum-classical image classification
(github.com/mujahidmahfuz/Hybrid-Quantum-Classical-Image-Classifier).

12. The Fujitsu quantum simulator is based on an ARM Architecture. Have the Member(s) used an ARM architecture before for testing the Problem set?

No ARM experience for this problem. Local development on x86 Linux (Ubuntu 22.04). All libraries (Qiskit, OR-Tools, NumPy) ARM-compatible.
Will Dockerize complete pipeline if needed.

13. List the libraries the Applicant(s) would require for using the quantum simulator.

1. Qiskit 1.1+ / Qulacs (QAOA circuits, QUBO solving). 2. QARP (Fujitsu-provided QAOA/VQE modules). 3. OR-Tools 9.8+ (classical VRP baseline).
4. NumPy, Pandas, NetworkX (QUBO matrices). 5. Folium, OSMnx (Tokyo route visualization). 6. Matplotlib, Seaborn (scaling plots).

14. If the libraries are not available, are the Applicant(s) willing to port them to the Fujitsu quantum simulator?

15. Where did you learn about Fujitsu Quantum Simulator Challenge 2025-26?

Fujitsu global technology page

16. Please make sure all Applicants' Member(s) have read and agreed to the Fujitsu Privacy Policy (Global) [<https://global.fujitsu/en-global/terms/privacy>] and the Privacy Notice, and then write a checkmark. ✓

Please note that participation in the challenge will require the Participant(s) to sign the Participant Agreement and Service Agreement that outlines participation conditions designated by Fujitsu. If the participant(s) will not sign these agreements, Participant cannot participate in the challenge.